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LOGICAL FICTIONS.

VI.

We have spoken, so far, of patterns and scales: we have done so only for the sake of convenience. It is impossible to find a useful pattern that is not scaled: if it were not scaled in some way there could be no pattern but only a symbol, as, for instance, *nothing* or *soul*: it is not possible to have a picture without some background. All, even our simplest patterns, are patterns standing out against a pattern which is, as it were, the background. "That" always implies the "not-that" from which it stands out.

We saw that this background is, if we go back far enough in our simplification, the infinite context; it is as if we could see a drop *in* the ocean, but as if it were separated, by ignoring the ocean. In practice the infinite context is "understood": a child's mother is seen against a limited background: quite early in life we get into the habit of using finite backgrounds—such as space and form scales. It is the purpose of this chapter to go more fully into the conception of *patterns based on other patterns*.

When we are conscious of the infinite context or of any part of it, there is a reaction, which it is for physiologists to explain. The reaction may or may not be "articulate": thus a pain in my toe may so affect me that I do not think: but if I do, I can put my thought into the form *that*, and also, probably *there*. This is very vague: but it is articulate—(even though I have no pain there). I have my

pattern *pain* and, though that pattern implies a non-pain background, I may be more interested in the space background *there* implying some knowledge of the vague background that is everywhere save there. The reaction, then, is usually a limited one.

It is important to recognize that, as we recognize a pain in the toe to be in us, so we must recognize that when we see the light of a new star we are seeing something that is as much *in us* as the pain. Also, that, as when I feel an itch on the tip of my nose I want to scratch it, so when I "see" anything I want to explain it.

Consciousness seems to be a reaction of a particular kind to the dominant part of a multiplicity of sensations—a momentary obsession. It is possible that what is called instinctive or purely physical reaction is more healthy than mental reaction. King Richard III may be more effective than King Richard II. We are only concerned with noticing that a murder and a name are both results of "wants" resulting from "obsessions." The cause of the obsession is what we have called a pattern: we pick it out from an indeterminate context: it becomes an obsession and we concentrate attention on it: and, as we do so, the pattern seems to crystallize, sending out rays or points in various directions, until the whole construction fades away under the influence of a new obsession.

Perhaps it will help if we think of a place where many roads meet. The actual meeting-place (what is called a Circus in well-known instances in London) will correspond to the nucleus of the pattern; given this central point, there will be vistas in various directions, all of which we might pass down; but as a rule we do not indulge in encyclopedic efforts of this kind: if "corn" is the "that" in question we shall pass either along to crops or the baker's shop—not to both. It will not be long before chance or choice has driven us through a series of obsessions, more or less con-

nected, with occasional bursts of what we call a train of thought. It will probably continue to be difficult to visualize the process until some one has invented some way of recording accurately the series of obsessions and reproducing them slowly. The thread a spider uses is only apparently a thread: so our thought is only apparently connected: given the means of magnifying sufficiently, we should probably be led to think of our mental universe as a body touched to sensitiveness in various parts—sensations flashing out here, there and everywhere in twinges. For it is probably much more than a simile to call a thought an ache.

The nearest we get to this desirable experiment is a baby thinking. It gazes—"yes," it thinks, "that is something: it's different from everything else." It then gazes at something else: and then looks back at number one. The difference here is fundamental: when it has only one pattern, it knows only "that" and the "not-that": but, having passed to the other pattern, it can ignore the very unsatisfactory "not-that" and play with two thats.

It would also be interesting to see, if thought could be slowed down for experimental purposes, whether we judge things, not, as is commonly supposed, entirely by what they are, but largely by what they are not. Thus if I see a car on a road, I do not really know where it is until I know (or have at least a sufficient idea of) where it is not. This is what one would expect, if a pattern is scaled: for a scale is useless unless we use it *as a whole*. The scale may be elastic, it may be vague; but we cannot know the intention of a scale unless we get its extension.

A few examples will serve to make this clear.

Timbuctoo: using the space scale we have it *there*: the scale is space, the world, the universe—as vaguely as may be, but none the less the whole space scale implied. Other scales such as time,—Timbuctoo when? now? or at some

other time?—or size (etc.) may also be implied. But it is clear that my knowledge of the position of Timbuctoo depends entirely on my knowing at least something about where it isn't.

Der Tag: This is a notorious pattern. But it is not hard to see that it is quite meaningless, unless we know about the days that were not "Der Tag." We cannot here go into an analysis of this phrase: but it would be a fascinating pursuit.

A bee: is it a peculiarity of my own mind that makes me suggest that in arriving at our idea of a bee we knock out certain things that it isn't? We see an instance of bee, I grant, without shuffling in this way; we either do or do not know what *bee* or the visible insect itself *stands for*. But what they stand for is not a mere proper name or label: in so far as it has meaning, we are bound to run along scales, fixing in our mind where *bee* comes by determining (as accurately as need be) where it isn't.

A Grain of Salt: a chemist clearly has a more or less clear idea of what it isn't. But if we consider the phrase as used in the sentence "to take a statement with a grain of salt," we have (have we not?) the same process. To take with a grain of salt is *not* disbelief, is *not* belief. The Japanese describe an argument about the size of a thing as *Okii chiüsai no arassi*—dispute of big? small? The phrase graphically describes the movement of the mind over a scale, until it comes to an intermediate position by excluding extremes.

Crimson: the simplicity and the complexity of such a pattern are evident: there is no meaning at all in the word—it is a mere label—unless we have some conception of the non-crimson; just as light-red can have meaning only as on a scale of all-the-reds. The more we know of what crimson isn't, the more accurately we know where crimson *comes on our scale*: what crimson is, is quite another matter.

Outrageous: this is a good example of definition by the negative. The word is equivalent to such a hypothetical word as "beyondageous": conduct is outrageous when it is *outré* or beyond every scale of conduct: such conduct is declared to find no place on the scale of conduct considered ethically: it is not good—it is not even bad—it is not even the worst conceivable—it is beyond anything that can be called conduct. It is a neat conception, similar to the very expressive Japanese *nai koto wa nai*, (isn't isn't, meaning "there is some").

Shrill as applied to sounds. We are apt to say that such qualities as shrillness are relative: so they are: if we are accustomed to a supershrill voice, we shall consider a rather shrill voice, perhaps, as not shrill at all. But if we knew no voices or noises that were not shrill, what meaning could we attach to shrill?

Elastic: that is a fascinating pattern: a child is quite at home with it, "knowing" all about what is not elastic: but as we come to consider what is not elastic, our conception of elasticity becomes more and more shaky: we must in order to understand elasticity understand non-elasticity—a task that becomes more and more difficult. For most of us elasticity will be understood only relatively in a small field of conditions. To avoid ambiguities, there ought to be as many words for "elasticity" as there are conceptions of the "non-elasticity" which scales it.

Death: this too is a pattern, the background or negative of which is now so vague that men can argue that there is nothing "alive" or that there is nothing "dead." But for practical purposes, the scale is got negatively: it is inconceivable apart from time and space scales: it is the absence of something (life) that has been somewhere: it is not mere darkness—it is darkness where light has been: just as when a child plays on the shore and makes sand pies with a pail—and the sand pie has the form of where

the pail wasn't,—so death is formed by the conception "life"—it is the pattern left by the removal of life. We cannot here go into this analysis more fully. But we have perhaps shown that here too our pattern (death) is meaningless unless we have hold of the pattern non-death which it implies and requires.

Matter: here we have a pattern of which the scale (or non-matter) has gradually been evaporated, so that it is possible to say that as (if) we can have no conception of what is non-matter, so we can have no conception of matter.

We have then in answer to the question "That is what?" a pattern which is (or may be) scaled. "That" is always like a silhouette which has no "meaning" except in so far as it stands out against a background. The wonderful rapidity with which we supply a series of backgrounds to our complicated conceptions gives them the appearance of body, life and movement—an illusion we may see on a very limited scale in any cinema. To some extent the process may be visualized (very incompletely and inadequately) as follows:

Take three points A, B and C, thus:

A

B

C

Time Scale: anywhere in the past (we imply that it is at some moment that could be but need not be stated).

Space Scale: somewhere in London (we could state the spot more accurately if necessary).

The spot A is a king one. Your mind is now active supplying a number of scales; by means of which your silhouette is gradually made to stand out until you can see a king. There can be no doubt

whatever that your silhouette and mine would be seen to differ if they could be recorded.

The spot B is a horse one—a white horse one—an Arab white horse one. Here again, you will admit, your mind has been hard at work: it has conceived and brought forth what you imagine I mean.

The spot C is a prince one. The same process produces a third silhouette.

All this is only preliminary to the real work: for these points are only circumstances or epithets qualifying what we are talking about—namely that there was then and there a giving which was a King—Prince—Horse giving, the direction being from the King to the Prince. Put in a formal way we are stating that One gave One One, A gave B to C, there was a B giving from A to C or an ABC giving then there. In this simple sentence “The King gave the prince a horse” we see that we have as our pattern a “giving” and that we give it meaning by an incredibly rapid use of a number of scales, by means of which we block in (as it were) a background until our pattern stands out like a snow peak against a cloudy background. Thus we get back to the iceberg we spoke of before: we deal with the visible portion—ignoring the infinite remainder, but using (as it were) the sea and the sky (these also having infinite remainders) as finite scales by means of which we outline that part of the iceberg which concerns us. It sounds complicated: but thought is complicated.

We must now deal with the difference between the questions “That is what?” and “That is who?” There is a subtle difference here: the failure to notice it (and it is a common failure) explains a host of mistakes in the past. The question “That is who?” posits individuality: and an individual is a pattern without a background or meaning.

As we saw before, an individual horse is usually called by anything but "horse." Any name does to mark an individual: in that an individual is unique, an individual is a class consisting of one member, and classification can go no further. Of course it is possible to consider a person not as a person but as a thing: instead of asking "Who is that?" it is possible to ask "What is that?" In early times there was a tendency to ask about mountains, trees, rivers, the sky, the earth and the stars "Who is that?" There is, throughout history a gradual reduction of the use of the question "Who is that?" Some of us may even have got so far as to ask "What is this?" for "What am I?" But it seems probable that until human passions and emotions are reduced to a state of impotence which many desire but few approach, we shall have to continue using the question "Who is that?" which implies individuality, a person, a soul or whatever it is we introduce into our thinking when we use a pattern without a scale. And until, as we said, the world changes so radically as to give up this interesting habit, taking to looking upon experience as full of what we have called scaled patterns, until that happens we shall continue to have three main divisions of philosophy: firstly, pure philosophy which is the proof that we can know nothing—given both in India and Greece long before the time of Christ; and then, applied philosophy which answers the two questions "That is what?" (and under this heading comes all science) and "That is who?" (and under this heading comes all ethical and religious speculation). We have no difficulty in admitting that, however complicated may be the process, we can get an answer to our question "That is what?"—a satisfactory answer, useful and fruitful—what we call SENSE. It may seem absurd, at first sight, to say that the answer to the question "That is who?" can never be sense: but not if we distinguish, as we shall have to, between nonsense (which has no meaning,

but has import) and bosh (which has neither meaning nor import). Our nurseries are full of the nonsense of religion: "*boys don't cry*" has no connection with the sense that has built up science; nor if small boys were looked upon as things would the phrase ever be used; but given the attitude implied in "my darling boy," what more natural, useful and important than such nonsense? Nonsense (not bosh) is the salt of thinking: the next chapter will deal with it more fully.

VII.

We have seen that to the question "What is that?" there can be no answer. To the question "That is what?" we answer in the form "That is that." Further than that we cannot go: the whole of science has been built up on that equation. But the students of pure philosophy who convince themselves that there can be no answer to the question "What is that?" and the students of science who are content to use only equations are still a minority in the world. What Professor Rhys Davids calls the "soul theory" is still prevalent; in olden days there was "an almost universal and unquestioned belief in the existence, round and about, of an infinite number of non-human beings. These the people took as a matter of course, just as they took the existence of souls inside their own bodies as a matter of course. It was by these souls, within them and without, that they explained to themselves the mysteries of death and trance, and dreams, of motion and of life." We still do. Our science is still shot through and through with traces of it. "Atoms" and the so-called "Laws of Nature" are often nothing but up-to-date "souls." Gautama seems to have tried to kill the soul theory: with what success we know. It is possible that man cannot thrive on nescience and equations: ages of superstition may have so tempered the human mind that it starves with-

out it: and perhaps if one tried to live without the soul theory he would go mad. This point (and many others) is fully dealt with in Burton's *Anatomy of Melancholy*. "*Dulce est desipere in loco*; to play the fool now and then, is not amiss; there is a time for all things." If pure philosophy teaches nescience, and applied philosophy the use of equations, it would seem that in order to keep the *mens hilaris*, the philosophers must be able to unbend and indulge in nonsense. And, though it is rightly to be called non-sense, being illogical or alogical, it will be best to use a less damaging name. Call the result of pure philosophy nescience and the results of applied philosophy either truth (which is merely the statement of an equation) or common sense (which differs from truth in that it postulates the theory of souls). Put more accurately: we have three main divisions of philosophy: (A) That part which deals with how and what we know: the answer being that we know by symbols, and that we know nothing, except (B) the equivalence of symbols, which is our science, the whole built up by elaborating the implications of equations; and (C) common sense which may be called an art built up in flagrant defiance of the first result of philosophy, namely nescience. Many philosophers have admitted the importance of the highly illogical products of common sense; others have despised them, as mathematicians may despise those who "waste their time" trying to square the circle. But what if it is natural to human beings (as natural as eating and breathing) to "try the impossible"? It seems not unreasonable to say that happiness is often found—perhaps always found—in those who are pursuing something that is obviously (even to themselves) out of reach. If that is so, our professors of philosophies will have in future to teach not only nescience, not only science, but also common sense.

If common sense is ever to become a part of philosophic

teaching (and we think it ought to) it is important to see exactly what it is and how it works. We have already seen that it is not the sense of science, which is always an equation on the model of "That is that." We have also seen that it seems to deal with patterns that cannot be scaled: for instance, the moment I look upon something as being somebody, I am lost to science: you can only introduce a somebody into an equation by turning him or her into an "it." If then our common sense has nothing to do with equations, how do we manipulate these not-to-be-scaled patterns?

The point happens to be brought out in the columns of *The Times* of London to-day. The Rev. Prof. C. F. Burney is arguing with the General of the Salvation Army about the truth of the Fall of Man and Original Sin. He says: "To suppose, however, that recognition of the mythological origin of the outward setting of these Hebrew narratives carries with it their 'scrapping' as the media of spiritual truth surely involves the crudest of misconceptions as to their purpose. The writers were actuated primarily, and mainly, not with the teaching of history or physical science, but with the teaching of *religion*. If once we lose sight of this fact, and think that we are to go to them for exact historical information, or for accurate scientific knowledge, we are certain to go wrong and to be disappointed. For it is in the sphere of religious truth that the inspiration of the Old Testament is to be found. The writing in which the truth is contained, whether it takes the form of history or any other form, is merely the human framework, and as such, subject to the limitations of human method."

This is not a remarkable piece of reasoning,—except perhaps for its inconsequence,—but it brings out very clearly the point that a positive value is attached to what is admitted to be scientifically untrue: it has value, in this

case, as religious truth. Now how can scientific untruth be in any sense truth?

Let us (ignoring the complication that is due to the use of the word *truth* in two contexts of this kind) take an illustration. If I want to start a horse I say "Gee! Up!" and in all probability the horse will move on. If I want to start a car, will "Gee! Up!" serve? Why not?

Now a car has no soul: we treat it scientifically: we don't (unless we lose our tempers) omit the scales by which we can foresee its reactions. We shall sooner or later get into serious trouble if we cannot name and scale various patterns inside the car. But the horse has a personality and, though we know something about horses, we are always up against something unique when we are dealing with one horse. It has been discovered empirically that a horse will react in certain ways to certain expletives. If we visit our horses by night, we know that it is wise to "say something" as we enter the stable to reassure the horses. We have, then, in addressing our cars no words; we act—if we can. In our intercourse with horses we act too—in so far as they are treated as machines: but we recognize the "soul" in them, and work on that with meaningless words. It will be admitted that we do much the same with young children: even with older children: even with adults—as, for instance, in times of war. We shout "The Hun is a pig" or "Allah is great" or "Death to the Foreign Devil": we do not claim to be "actuated primarily, and mainly, with the teaching of history or physical science, but with the teaching of" patriotism. Unfortunately this claim that we don't mean what we seem to mean is a very dangerous one. If I could persuade your horse that when you say "Gee! Up!" you don't mean anything, he might learn to pay no attention. Expletives are useless unless the target at which they are aimed reacts to them: and inanimate (or soulless) targets cannot thus react. Hence

the first condition: expletives (or common sense) can only be used with individuals (souls). The second condition is: expletives need mean nothing, or they may seem to mean the opposite of what they say: but expletives are useless unless the soul aimed at reacts to the expletive in the desired way.

We have used the term "expletive" purposely: it evidently covers only a very small portion of what is meant by common sense: but it serves to emphasize the very great difference between the language that is used for persuasion and small talk and propaganda of all kinds and the language of science: it also serves to remind us that we have much more to do with "souls" than with science: and the tact which suggests to a man the right expletives to use in any given situation is rightly considered a very valuable asset. Much has already been done to reduce the art of using expletives to a science (crowd management, factory management, the mind of the heathen and so on). It is a complicated question: we have to discover what noises will produce desired effects under given circumstances: and a study of the expletives used during the last general elections in England makes it quite clear that the personality of the speaker must have something to do with it too: it is not right to say that the public will swallow anything: every public requires the right expletives from the right fellow at the right moment. Given those, expletives are indeed (as Kipling once said when speaking of those expletive manufacturers usually called poets) words that walk up and down in the heart of men.

We suggested that this might be made a part of the philosophy teaching of universities. But only in this way: before a degree or any kind of certificate of proficiency for philosophy be awarded, the professors or examiners or both should have satisfied themselves that the candidate has had practice in the use of nonsense, common sense,

expletives or whatever else we choose to call this very human, very important and very unscientific procedure.

Philosophy is an exacting study: it is apt to cause melancholy: hence it should be not only a maxim, but a condition of qualification that a student should not fail "amidst his serious studies and business, to use jests and conceits, playes and toyes, and whatsoever else may recreate his mind." Intercourse with fellow beings or with animals, not theory, teaches how to use words in this meaningless but very effective way. It is far too often assumed that language is, as a rule, used scientifically and grammatically: there are languages that are, as a rule, used thus; for instance the language of mathematics, the language of chemistry; and the notations of music and of phonetics and certain conventions of commerce are accurate. But for the most part words, even in much that claims to be science, are (like an otherwise useless kind of egg) for electioneering purposes. It is the rarest thing to find any one *convinced* in a debate: we start an argument and are not in the least surprised if both sides retire with the conviction that the other side has been talking nonsense: this may very well be so: but our purpose sometimes is to arrive if possible at a conclusion, which is not to be expected unless we start with "given this and that, let us see what follows." Far too often we start from different points of view: the result is as satisfactory as if you and I sat down to play a game, you with chessmen and I with draughts. The trouble is largely due to a failure to distinguish between these three kinds of "knowledge"; in the next section we shall attempt to distinguish between the products of "knowledge," which are usually all grouped under one name "truth," which, in consequence, is an ambiguous term. Let us call the knowledge of pure philosophy K_1 ; of science (i. e., equations) K_2 ; of common sense

(efficient expletives) K₃; and let us see how the "truth" derived from them differs.

VIII.

Bishop Berkeley says that the physical universe which is seen and felt and inferred is just a dream—your dream and mine and nothing else; only it so happens, he says, that our dreams agree in many respects. It is natural to inquire how it "so happens."

We have already seen that in practice we deal with three kinds of so-called knowledge: the fact that one name is applied indiscriminately to all three, has been the source of much misunderstanding. There is, firstly, the knowledge that all knowledge is of symbols, these being the tangible signs which stand for a fictitious relation between otherwise incomprehensible variables—a relation that is assumed to be constant until the deductions from such an assumption are seen to be unsatisfactory. This is K₁. There is, secondly, the knowledge we gain by assuming the equivalence of symbols as if they were constants; by ignoring the variables in the equation as negligible quantities we get what is in theory highly illogical but in practice most useful and fruitful—a fictitious decomposition of the infinite. This is K₂. There is, thirdly, the knowledge which we gain by emphasizing the variability of the values of symbols (positing, in fact, individuality) and treating them as a man aims at a moving target. Using the words strictly, this knowledge is instinctive and emotional—K₃.

Now if in Pure Philosophy there is no truth, if the burden of it is that we can know nothing, it may be objected that if we know nothing, we cannot know that we know nothing. That this objection should so often be a stumbling block is due to an impetuous belief that language will bear any weight: like children skating, we may forget that there are spots which will not bear.

What is meant by saying (idiomatically) that we know nothing is similar to what is meant when we say (again idiomatically) that "there is nothing in that box." This statement, taken literally (as the objectors to our original statement take ours), is absurd: a box with *nothing* in it is not what we really mean: we mean that it is a box with *something else* in it—not the something we were thinking of. Likewise, if I say I have no knowledge of things—that I know no thing, I do not mean that I am daft, but that what I know is *no thing* but some relation. I skate over what is like a frozen surface with great ease and profit: language is, as it were, the ice: you may emphasize, if you will, that language represents reality, as ice "represents" water: no one can reasonably deny that, as we have a body of water under a frozen surface, so we have "reality" behind language: at the same time if you fall in—that is, if the ice gives way—your skating is over and it is a matter of saving your soul alive: so if language gives way and we fall into "reality," our thinking is over and it is a matter of saving our sanity.

Or, to put it another way: I may know nothing about the contents of two bags: but I may know how they behave on the scales: if one is heavier than the other, I can discover what should be added to the one to make it equal to the other, and so on. Words (or the patterns they present) are like such bags: we can do great business with them: but we can never open them: in fact it is only the over curious who ever wonder what is inside them: they cease, in practice to be bags at all, they become weights, and most of us would answer, like a trader asked what was in his pound weight, "why, it's solid."

If, then, of truth there are three kinds, let us use three distinct symbols: the three kinds of knowledge will produce three different results.

Let us keep the word TRUTH as the name of the product

of K₂. As the product of K₁ is a less generally useful kind of truth, let us lop off the first letter and call it RUTH. And as K₃ produces a kind of truth that posits the soul, or personality, let us add one letter (a relic of an old expletive) and call it STRUTH.

RUTH, the product of K₁, must have taken many centuries of thought: we find it complete in the Upanishads. It is reached, like truth, by the gradual elimination of the self, the soul, the personality, until we get such expressions as the Impersonal Self. If we express it mathematically it is the equation

$$\begin{aligned}\text{Everything} &= \text{nothing} \\ \text{all} &= 0.\end{aligned}$$

It looks useless enough: but it shuts out a number of perplexing problems. That equation is to us, what the bars of the cage are for wild beasts in the Zoo. We take this as granted: we become reconciled: we recognize our bearings. We shut our eyes to the discrepancy that must always exist between *this all* and *the all*, and we assume that we may, for instance, say

$$\begin{aligned}\text{this all} &= \text{all that} \\ \text{and disregarding the fact that all} &= 0, \text{ we continue} \\ \text{this} &= \text{that.}\end{aligned}$$

On this equation (and this assumption) is built up the whole of science, or what we have called TRUTH.

But scientists are often human: we find them using, for instance, such phrases as "*E pur se muove.*" This is not merely a statement of truth: the author is betraying certain διαθέσεις ψυχῆς; the phrase may contain truth: but it also contains something of the nature of an expletive, something of the emotional attitude that introduces the name of God; we have, in fact, a clear streak of STRUTH as well as TRUTH.

Our ignorance of the origins of language is immense;

but if we keep clearly before us the difference between the pattern (or the "thought") and the label (or the expression) we are tempted to suppose that in the most rudimentary stages the pattern was highly individual and personal, and the expression purely expletive. A dog's patterns must represent a very advanced stage of composition and intellectual comprehension. We may reasonably imagine a dog being continually in the attitude of asking "Who is that smell?" or "Who is that noise?" or "Who is that patch?" The expression of his thoughts, which may be too subtle for us to analyze, seems monotonous. In other words, the dog has a "language" that is highly ambiguous. We are told that in Brazilian the one word *tuba* may mean his father, or he has or he is a father, as if this were a peculiarity of a primitive language. We might as well conclude that French is a rudimentary language because the word *à* requires no less than twelve columns in Littré's dictionary. It is a very bad (but very natural) mistake to suppose that because an expression looks awkward to us, it *is* awkward. The phrase "rudimentary language" is useless. We have to deal, on the one hand, with patterns of *thought*: these we call rudimentary if they strike us as inferior to our idea of highly developed patterns of thought. It is open to Gough to say: "the Upanishads are the loftiest utterances of Indian intelligence. They are the work of a rude age, a deteriorated race, and a barbarous and unprogressive community."¹ My dog has every right to say the same of my work, and he could say it in one word. On the other hand we have *expression* which is more or less convenient (and convenience is largely a matter of habit): the Greek particles are a curse to some minds, the Russian verbs a puzzle to others, such phrases as the Japanese: *Inu to iu mono wa*, or the French: *qu'est-ce que c'est?* may seem to foreigners absurdly cumbrous; on the other hand, a

¹ A. E. Gough, *The Philosophy of the Upanishads*, 1882, p. 268.

himalayan amount of meaning is often carried by one word in all languages: it is only relatively convenient to use such expressions as *sour grapes* or as the Japanese proverb *Hotoke no kao mo san-do*, which Chamberlain translates by "Even a Buddha's face can only be tickled thrice" (or the crushed worm will turn). It is not necessary to multiply examples. Expression is clearly convenient, good, beautiful, highly developed and so on, only if it happens to take the judge's fancy. But thought and expression may always be classified as colored or colorless. "Color" is the result of the intrusion of personality. This subtle change, as if the sun suddenly shone on the words, may be seen as follows: in languages (such as Japanese) that have no relatives we get phrases like "*Yesterday came man,*" for the thought we express by "*the man who came yesterday.*" At present this phrase is colorless: it is a term that might enter into an equation. But if we invert and, instead of saying "*the man who came yesterday,*" say "*who came yesterday? —the man*" (this is perhaps the earliest form and origin of the relative) we at once introduce the presence of a speaker and a hearer. When a child says "*Three sevens are twenty-one*" we call it truth. When it says "*Three sevens are twenty-one, aren't they?*" we have that perplexing expletive, which turns colorless truth into a highly colored specimen of struth: and the answer whatever it is (unless the child happens to have an exceptional parent) will mean something like "*I think that is what we said before.*"

It is perhaps impossible to prevent the language that is in common use from being colored. If the Buddha and Socrates had invented new symbols for their thoughts instead of using chameleonlike symbols which everybody thought he understood, we might now be able to write the history of two sciences, instead of having to write the history of two religions. Geometry has been made possible

because (though even here color has crept in) circles and points were called A, B, C, or D. All serious sciences have had to fall back on a specialized form of expression which gave an impersonal quality (as far as this can be done). How dangerous ordinary language is may be seen if we consider a common opening to mathematical expositions: "Let. . . ."—here is already a flavor of struthfulness: all imperatives presuppose two (at least) persons—like our expletive "Gee! up!" It seems likely that, in early forms of speech, there were words like "up" that stood for a command when the observer was feeling aggressive or a noun when the aggressor was feeling observant, as for instance in face of a steep hill which he wished to get to the top of. Be that as it may, our scientist would be well advised to leave out even such apparently colorless words as *if* (*given*) and express in a strictly colorless symbolism his successive steps

$$(1) \quad a = 3b$$

$$(2) \quad a = 2b + b$$

$$(3) \quad a - b = 2b$$

To sum up: there are three kinds of truth and three kinds of knowledge. What we have called TRUTH is got by equations and the elimination of personality. There are two ways of eliminating the color of personality (1) by turning out the light altogether and getting darkness: what we have called KI and RUTH; (2) by getting a pure white light (as it were by fusing all personalities into impersonality) which may indeed (either intentionally or unintentionally) be split up, but which, the moment it shows traces of the color of personality, ceases to be TRUTH and becomes what is, scientifically considered, nonsense; socially considered, that very valuable and illogical kind of logic known as common sense or, as we prefer to call it, STRUTH.

IX.

If we drive analysis right home, pure philosophy leads to nescience and everyday language to non-sense. The language of science produces sense by the use of equations based on the fundamental assumption that we are justified in saying *this* = *this*. It must not be supposed that this is anything more than an assumption that is justified only because it is, within its limits, true enough. Much confusion is bound to arise the moment we assume that *this* is indeed equal to *this*. We have seen that every *this* has an infinite remainder which we can, up to a certain point, afford to ignore: it is therefore clear that, even in the department of knowledge to which we have limited the use of the term "truth," we are able to proceed only because we refuse to recognize, on either side of the equation, the presence of immeasurable variables. The equation should be *all this* = *all this*: and it is by dropping the *all*, with which we cannot deal, that we can conveniently handle the *this*.

If I say that water boils at such and such a point, and always boil my water in the same room, I may not discover that this truth is only true enough; but, if I boil my water high up in the mountains, I shall find errors in my calculation, because part of the infinite remainder has become relevant, showing that our *this* = *this* is no longer true, because our *this* was too strictly defined.

In the same way it has been assumed that time and space could be treated always as if 1 hour = 1 hour or 1 yard = 1 yard. Within limits—and the limits in this case have only recently been reached—these assumptions are true enough. But it is often forgotten that they are assumptions which can only be treated as absolutely true for the sake of convenience.

The statement that this is not really equal to this is described sometimes as outside common sense. This is

precisely what it isn't. As we have seen it is science and not common sense that affirms that *this man* is the same as *this man*. The statement is a palpable fiction, which common sense is far too wary to assume: it is much safer in practical life to emphasize the variables and to act on the principle that *this* may have been a very good boy half a minute ago, but there's no telling what *this* may be in half a minute's time. Ladies, too, know quite well that a dress in a sense remains *the same*: but is it the same on her? Two men use the same words: are they the same? It is amusing to find commonsense accusing science of talking non-sense when science has for a moment allowed common sense to intrude into the field of fiction in which it is wont to work.

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